

**Response to**

Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

**Agency**

Networking and Information Technology Research and Development (NITRD) National  
Coordination Office (NCO), National Science Foundation.

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This document is approved for public dissemination

## **Why Agriculture?**

United States Agriculture is poised to be the global leader in food production through the implementation of agriculture technology (“AgTech”) and artificial intelligence. The agriculture industry is considered to be top industry to be impacted by artificial intelligence in 2025, and is highly strategic to the United State’s national security (Purnama & Sejati, 2023; Spanaki et al., 2022). Food security is often labeled as national security. Global population is expected to increase to nearly 10 Billion people by the year 2050, that is a 30% increase (Van et al., 2021). To support this population growth, it is expected the agriculture industry will need to produce nearly 100% more food.

During a presentation by John Deere’s (Headquarters in Moline, IL) Chief Technology Office on February 27<sup>th</sup> at Grand Farm’s Innovation Campus in Casselton, North Dakota – the potential and impact of artificial intelligence was woven throughout the keynote. Advancements in AI have produced exponential increases in capabilities of intelligent machinery. The federal government has the potential for targeted investments, grants, and collaborations for this potential to be realized.

## **The Challenge of Artificial Intelligence in Agriculture**

Grand Farm has been working with thought leaders and executives for America’s largest agriculture and technology corporations, and research leaders from around the country to understand the role of the United States Government (through the United States Department of Agriculture) in accelerating artificial intelligence research and development and the adoption of these tools into the agriculture industry. Through these interviews and focus groups, several themes have emerged:

1. Aligning AI innovation with agricultural stakeholders
2. Lowering barriers to AI adoption in agriculture
3. Creating a secure, accessible, and interoperable agricultural data ecosystem
4. Identifying challenges to address that are relevant to United States farmers

Agriculture has significant challenges to implementing artificial intelligence innovations. First, agriculture technology is reliant on near-peer innovation. To implement artificial intelligence, you need infrastructure for both last-mile and last-acre connectivity, the sensors that can collect data on the field and crops, data centers for cloud computing, and algorithms to produce decisions with that data that are actionable for farmers. Each of these cannot exist without the other. Second, agriculture is uniquely challenged by the uniqueness of each individual farm. Artificial intelligence relies on patterns to create models, and immense amounts of data to improve the accuracy of those models. Each farm has different soil conditions, weather, crops they grow, and how the business of farming is conducted. Both of these challenges will require public-private collaboration to overcome.

## **The Solution:**

Stakeholder engagement is immensely important in defining key opportunities and challenges that artificial intelligence can solve. Agriculture is unique in that each geography (and each farm) have unique perspectives on what is needed. The challenges of the upper-midwestern United State grower are different than those of the south-east. This will need to be regularly conducted as the challenges and opportunities also regularly change year-over-year.

Over the past several years, innovation ecosystems have seen substantial investments from the federal government. This has achieved amazing results in removing barriers, increasing acceleration of these technologies, and driving adoption into American society. This is an area to double-down, finding key ecosystem facilitators to work on these immense challenges and driving forward successful solutions.

Similar to ecosystems, research and development test beds have also seen investment. As the challenges of technology development and adoption continue to get more complex, these test beds provide a rapid way to demonstrate, research, develop, and validate technology. By having test beds intentionally designed for these activities, the cost on startups and private industry to develop these technologies dramatically decreases.

Grand Farm is uniquely positioned to engage in both ecosystem building and test bed activities. As a core member of the NSF FARMS: North Dakota Advanced Agriculture Technology Engine, we have been able to grow our global innovation ecosystem to over 2,500 organizations. These organizations have been mobilized to solve challenges and opportunities unique to United States agriculture. Through this effort Grand Farm has also engaged with growers and commodity leaders in defining opportunities and challenges to be solved by this global ecosystem through careful and thoughtful stakeholder analysis. As a consortium member of Economic Development Administration's Technology Hubs, Grand Farm has developed a test bed at Grand Farm's Innovation Campus to provide an environment for rapid prototyping, testing, and validation of technology. As a partner of the United States Department of Agriculture's Agricultural Research Service, we have contributed to the artificial intelligence strategy of USDA. Through partnerships with both North Dakota State University and University of Georgia, Grand Farm has been able to work on regional opportunities and challenges unique to their growers – with expansions planned into other areas of the country in 2025 and 2026.

We have until 2050 to double our food production globally. The United States is and will remain the leader in this effort with intentional investments into artificial intelligence. We look forward to working with the National Science Foundation to successfully meeting this challenge head-on.

**References:**

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